# Sahaj Bhadja

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## Summary

Full-stack platform engineer experienced in **React, TypeScript, Python, Java, APIs, distributed systems, data pipelines, and AI applications**, building scalable, reliable services and developer-friendly workflows for data-rich products.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
**Coursework:** Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
**Coursework:** Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **LA, California**
*Software Engineer* *September 2025 – Present*
- Built scalable FastAPI platform services processing 5,000+ monthly data jobs with explicit state transitions, failure isolation, validation, and reconciliation, increasing concurrent throughput 3x while achieving 99%+ reliability.
- Designed multi-tenant PostgreSQL and SQL Server storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through normalized schemas, indexing, caching, and query optimization.

**Grad8** **LA, California**
*Software Engineer Intern* *June 2025 – August 2025*
- Shipped React and Flask capabilities integrating multimodal AI models, OCR, AWS RDS, and REST APIs, supporting 150+ instructors and efficiently processing 10,000+ daily requests with reliable performance at scale.
- Implemented reusable backend modules and concurrent ingestion for 1,000+ files per batch, adding automated tests, diagnostics, fallback paths, and cost tracking while reducing feedback turnaround time 60%.

**USC AD Labs** **Los Angeles**
*ML Research Assistant* *January 2025 – May 2025*
- Developed Python and PyTorch components processing 100,000+ trajectory records, reliably improving long-horizon planning 18% through rapid prototyping, simulation, debugging, reproducible testing, and iterative model integration at scale.
- Collaborated with senior researchers to analyze edge cases and system interactions, increasing closed-loop stability 15% while documenting architecture tradeoffs, experimental findings, and reproducible implementation decisions at scale.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Developed cross-platform Java and Python microservices using REST APIs, Docker, SQL, AWS Glue, and S3, improving response times 30%, reducing payload sizes 25%, and lowering storage costs 15%.
- Owned requirements-through-deployment platform improvements, automating 20+ tests across Windows, macOS, and Linux while increasing pipeline efficiency 68% and collaborating with Microsoft and GitHub stakeholders during Agile reviews.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Built React and Node.js capabilities across eight reusable microservices, reliably synchronizing two million records through REST APIs, WebSockets, Azure SQL, authentication, resource allocation, billing, and notification workflows.
- Automated testing and Docker-based deployments through Jenkins, reducing release time 50% while using code reviews, monitoring, and operational analytics to improve platform quality across production release cycles.

## Skills

**Languages**: Python, Java, TypeScript, JavaScript, SQL, C++, Shell, Bash
**Full Stack/AI**: React, Next.js, Node.js, FastAPI, Flask, REST APIs, WebSockets, PyTorch, LLMs
**Data/Platform**: PostgreSQL, MongoDB, distributed systems, microservices, Docker, Kubernetes, AWS, Azure, GCP
**Quality**: Git, CI/CD, Jenkins, PyTest, Jest, unit testing, integration testing, debugging, monitoring

## Projects

**Big-Brain** Live | *Enterprise AI Platform*
- Built AI-powered retrieval over 50,000+ documents using PostgreSQL, GCP, REST APIs, LLaMA-2, FAISS, and Pinecone, achieving sub-second search while reducing retrieval latency 40% through scalable pipeline optimization.

**Chat2SQL** Code | *Developer Tool, Java, TypeScript*
- Developed a modern AI-powered developer workspace across Java and TypeScript services, supporting 10+ databases with schema-aware query generation, structured validation, Docker workflows, and reliable cross-platform local deployment.

**PiXMod** Live | *Full-Stack AI SaaS*
- Built a Next.js AI platform integrating CLIP, SAM, Stable Diffusion, and MongoDB, processing 100,000+ image embeddings while delivering semantic search, generative editing, scalable storage, and subscription workflows.